

LEVEL - VI**SINGLE ANSWER QUESTIONS**

1. If $f(x) = |x-2|$ and $g(x) = f(f(x))$, then $\sum_{r=0}^3 g'(2r-1)$ is equal to
A) 0 B) 1 C) -1 D) 2
2. Let $g(x)$ be a polynomial of degree 1 and $f(x)$ be defined by

$$f(x) = \begin{cases} g(x), & x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}}, & x > 0 \end{cases}$$
 If $f(x)$ is continuous such that $f'(1) = f'(-1)$, then $\frac{g(x)}{x} =$
 A) $-\frac{2}{3} \ln \frac{3}{2} + \frac{1}{9}$ B) $\frac{2}{3} \ln \frac{3}{2} - \frac{1}{9}$
 C) $\frac{2}{3} \ln \frac{3}{2} + \frac{1}{9}$ D) $-\frac{2}{3} \ln \frac{3}{2} - \frac{1}{9}$
3. If $(x+1)(x+2)\dots(x+n) = A_0 + A_1x + \dots + A_nx^n$ then $A_1 + 2A_2 + \dots + nA_n =$
 A) $(n+1)! \left[\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1} \right]$
 B) $n! \left[\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1} \right]$
 C) $(n+1)! \left[1 + \frac{1}{2} + \dots + \frac{1}{n+1} \right]$
 D) $n! \left[1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1} \right]$
4. If $f(x) = \sqrt{x+2\sqrt{2x-4}} + \sqrt{x-2\sqrt{2x-4}}$, then $f'(3) + f'(6) =$

A) 0 B) 1 C) 2 D) $\frac{1}{2}$

5. If $y = 2 + |x| - |x-1| - |x+1|$, then $y'\left(-\frac{1}{2}\right) + y'\left(\frac{1}{2}\right) + y'\left(\frac{3}{2}\right) + y'\left(\frac{5}{2}\right) =$
 A) 0 B) 1 C) -1 D) -2
6. Let $f: R \rightarrow R$ be a differentiable function such that $f(2) = -40, f'(2) = -5$ then

$$\lim_{x \rightarrow 0} \left(\frac{f(2-x^2)}{f(2)} \right)^{\frac{4}{x^2}}$$
 is equal to
A) e^{32} B) $\sqrt{e}$ C) $\frac{1}{\sqrt{e}}$ D) e

7. If $y = \log_u (|\cos 4x| + |\sin x|)$ where

$$u = \sec 2x \text{ then } \left(\frac{dy}{dx} \right)_{x=-\frac{\pi}{6}} \text{ is}$$
A) $\frac{-\sqrt{3}(12 + \ln 2)}{\ln 4}$ B) $\frac{3\sqrt{3}}{\ln 4}$ C) 0 D) $\frac{1}{2}$

8. Let $\frac{f(x+y) - f(x)}{2} = \frac{f(y) - a}{2} + xy$ for all real x and y . If $f(x)$ is differentiable and $f'(0)$ exists for all real permissible values of a and is equal to $\sqrt{5a-1-a^2}$ then $f(x)$ is
 A) positive for all real x
 B) negative for all real x
 C) $f(x) = 0$ has real roots
 D) cannot say anything about the sign of $f(x)$

9. If $R = \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{3}{2}}}{\frac{d^2y}{dx^2}}$, then $R^{\frac{2}{3}}$ can be put in the form of

$$A) \frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} + \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}}$$

$$B) \frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{3}{2}}} + \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{3}{2}}}$$

$$C) \frac{2}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} + \frac{2}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}}$$

$$D) \frac{1}{\left(\frac{d^2y}{dx^2}\right)^{\frac{2}{3}}} \cdot \frac{1}{\left(\frac{d^2x}{dy^2}\right)^{\frac{2}{3}}}$$

MULTI ANSWER QUESTIONS

10. If $f(x) = \sin^{-1}(\sin x)$, then $f'(x)$ is

A) 1 for all x
 B) -1 for all x
 C) -1 in 2nd and 3rd quadrant
 D) 1 in 4th quadrant

11. If $y = \frac{\sqrt{1+t^2} - \sqrt{1-t^2}}{\sqrt{1+t^2} + \sqrt{1-t^2}}$ and $x = \sqrt{1-t^4}$ then $\frac{dy}{dx}$ is

$$A) \frac{-1}{t^2(1+\sqrt{1-t^4})} \quad B) \frac{(\sqrt{1-t^4}-1)}{t^6}$$

$$C) \frac{1}{t^2(1+\sqrt{1-t^4})} \quad D) \frac{1-\sqrt{1-t^4}}{t^6}$$

STATEMENT TYPE QUESTIONS

12. Statement 1 : For

$$f(x) = \sin x, f'(\pi) = f'(3\pi)$$

Statement 2 : For

$$f(x) = \sin x, f(\pi) = f(3\pi)$$

A) If both the statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1
 B) If both the statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1

C) If STATEMENT 1 is true and

STATEMENT 2 is false

D) If STATEMENT 1 is false and

STATEMENT 2 is true

13. Consider function $f(x)$ satisfies the

relation, $f(x+y^3) = f(x) + f(y^3), \forall x, y \in R$ and differentiable for all x .

Statement 1 : If $f'(2) = a$ then $f'(-2) = a$

Statement 2 : $f(x)$ is an odd function.

A) If both the statements are TRUE and STATEMENT 2 is the correct explanation of STATEMENT 1

B) If both the statements are TRUE but STATEMENT 2 is not the correct explanation of STATEMENT 1

C) If STATEMENT 1 is true and

STATEMENT 2 is false

D) If STATEMENT 1 is false and

STATEMENT 2 is true

COMPREHENSION TYPE QUESTIONS

If equation $f(x) = 0$, where $f(x)$ is a polynomial function, and if it has roots $\alpha, \alpha, \beta, \dots$ then $f(x) = 0$ is equivalent to

$$k(x-\alpha)^2(x-\beta)\dots = 0, (k \neq 0) \text{ from}$$

which we can conclude that $f'(x) = 0$

$$\text{or } 2(x-\alpha)[(x-\beta)\dots] + (x-\alpha)^2[(x-\beta)\dots]' = 0$$

$$\text{or } (x-\alpha)[2(x-\beta)\dots + (x-\alpha)[(x-\beta)\dots]'] = 0$$

has root α . Thus if α root occurs twice in equation, then it is common in equations

$f(x) = 0$ and $f'(x) = 0$ similarly, if α root occurs twice in equation, then it is common in the equations

$$f(x) = 0, f'(x) = 0 \text{ and } f''(x) = 0$$

14. If $(x-c)$ is a factor of order m of the polynomial $f(x)$ of degree n ($1 < m < n$) then $x=c$ is a root of the polynomial (where $f^r(x)$ represent r th derivative of $f(x)$ w.r.t x)

DIFFERENTIATION

- A) $f^m(x)$ B) $f^{m-1}(x)$
C) $f^n(x)$ D) $f'(x)$

15. If $a_1x^3 + b_1x^2 + c_1x + d_1 = 0$ and $a_2x^3 + b_2x^2 + c_2x + d_2 = 0$ have a pair of repeated roots common then

$$\begin{vmatrix} 3a_1 & 2b_1 & c_1 \\ 3a_2 & 2b_2 & c_2 \\ a_2b_1 - a_1b_2 & c_1a_2 - c_2a_1 & d_1a_2 - d_2a_1 \end{vmatrix}$$

- A) 0 B) 1 C) 2 D) 3

16. If α roots occurs p times and β root occurs q times in polynomial equation $f(x) = 0$ or n degree ($1 < p, q < n$) then which of the following is not true. (where $f^r(x)$ represents r th derivative of $f(x)$ w.r.t x)

- A) If $p < q < n$ then α & β are two of the roots of the equation $f^{p-1}(x) = 0$
B) If $q < p < n$, then α & β are two of the roots of the equation $f^{p-1}(x) = 0$
C) If $p < q < n$ then equations $f(x) = 0$ and $f^p(x) = 0$ have exactly one root common
D) If $q < p < n$ then equations $f^q(x) = 0$ and $f^p(x) = 0$ have exactly two roots common

MATRIX MATCHING QUESTIONS

17. COLUMN - I

A) $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$, then $\frac{dy}{dx} = -\frac{2}{1+x^2}$

B) $y = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)$, then $\frac{dy}{dx} = \frac{-1}{1+x^2}$

C) $y = |e^{|x|} - e|$, then $\frac{dy}{dx} > 0$

D) $u = \log|2x|$, $v = |\tan^{-1} x|$, then $du/dv > 2$

COLOUMN - II

- p) for $x < 0$
q) for $x > 1$

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- r) for $x < -1$
s) for $-1 < x < 0$

INTEGER TYPE QUESTIONS

18. If $y = SX$ and $z = tX$ where all the letters denotes the functions of x and suffixes denotes the differentiation w.r.t x and

$$\begin{vmatrix} X & Y & Z \\ X_1 & Y_1 & Z_1 \\ X_2 & Y_2 & Z_2 \end{vmatrix} = X^K \begin{vmatrix} S_1 & t_1 \\ S_2 & t_2 \end{vmatrix} \text{ then } K \text{ is}$$

19. If $f(x) = \prod_{n=1}^{100} (x-n)^{n(101-n)}$ and

$$\frac{f(101)}{f'(101)} = \frac{k}{5050} \text{ then find } k.$$

20. Let $g(x) = \log(f(x))$ where $f(x)$ is twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = xf(x)$ then for

$$N=1,2,3,\dots \text{ and } g''\left(N+\frac{1}{2}\right) - g''\left(\frac{1}{2}\right) =$$

$$k\left(1+\frac{1}{9}+\frac{1}{25}+\dots+\frac{1}{(2N-1)^2}\right) \text{ then 'k' is.}$$

21. If the function

$$f(x) = -4e^{\frac{1-x}{2}} + 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$$

$$\text{and } g(x) = f^{-1}(x) \text{ and } g'\left(\frac{-7}{6}\right) = \frac{1}{k}$$

then 'k' is.

22. Let $f(x) = (x-1)(x-2)\dots(x-n)$, $n \in N$ and $f'(n) = 5040$ then $n =$

- 23.

$$\text{Let } f(x) = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$$

then $\sqrt{\frac{f(50)f'(50)}{2}} =$

24. Let $F(x) = f(x) \cdot g(x) \cdot h(x) \forall x \in R$, where $f(x)$, $g(x)$ & $h(x)$ are differentiable function. At some point x_0 , $F'(x_0) = 21 F(x_0)$,

$$f'(x_0) = 4f(x_0), g'(x_0) = -7g(x_0), h'(x_0)$$

$$= k \cdot h(x_0) \text{ then } \frac{k}{4} =$$

25. The function f satisfies the relation $f(x) \cdot f'(-x) = f(-x) f'(x) \forall x$ & $f(0) = 3$, Now if $f(3) = 3$ then $f(3) \cdot f(-3) =$

26. By using the substitution

$$Z = \ln\left(\tan \frac{x}{2}\right) \text{ the equation}$$

$$\frac{d^2y}{dx^2} + (\cot x) \frac{dy}{dx} + 4y \sec^2 x = 0$$

transforms to $\frac{d^2y}{dz^2} + Ky = 0$ then K is

27. Let $y = x \sin Kx$ then sum of all possible values of K such that

$$\frac{d^2y}{dx^2} + y = 2K \cos Kx \text{ holds true for all}$$

$$x \in R \text{ is}$$

28. Let f is a real valued function defined on the interval $(-1, 1)$ such that

$$e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1} dt \quad \forall x \in (-1, 1)$$

and let f^{-1} is the inverse of f . If

$$(f^{-1}(2))' = \frac{1}{k} \text{ then } k \text{ is}$$

SUBJECTIVE TYPE QUESTIONS

29. Find the derivative of

$$f(x) = \begin{cases} \frac{x-1}{2x^2-7x+5} & \text{when } x \neq 1 \\ -\frac{1}{3} & \text{when } x = 1 \end{cases}$$

30. If $y = \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{c}{x-c} + 1$

prove that $\frac{y'}{y} = \frac{1}{x} \left(\frac{a}{a-x} + \frac{b}{b-x} + \frac{c}{c-x} \right)$

KEY LEVEL - VI

- | | | | |
|---------------------------------------|----------|----------|--------------------|
| 1) A | 2) D | 3) A | 4) D |
| 5) D | 6) C | 7) B | 8) A |
| 9) A | 10) C, D | 11) A, B | 12) B |
| 13) A | 14) B | 15) A | 16) D |
| 17) A-q, r ; B-p, r ; C-q, s ; D-q, r | | | |
| 18) 3 | 19) 1 | 20) 4 | 21) 5 |
| 22) 8 | 23) 5 | 24) 6 | 25) 9 |
| 26) 4 | 27) 0 | 28) 3 | 29) $-\frac{2}{9}$ |

LEVEL - VI HINTS

SINGLE ANSWER QUESTIONS

$$1. \quad g(x) = \begin{cases} |x|, & x \leq 2 \\ 4-x, & 2 < x \leq 4, \\ x-4, & 4 < x \end{cases}$$

$$g'(-1) + g'(1) + g'(3) + g'(5) = -1 + 1 - 1 + 1 = 0$$

$$2. \quad \text{Let } g(x) = ax + b$$

$$f \text{ is continuous} \Rightarrow f(0) = f(0^-) = f(0^+) = 0$$

$$b = \lim_{x \rightarrow 0} \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} = 0 \text{ since } \left(\frac{1}{2} \right)^{\infty} = 0$$

$$\frac{d}{dx} \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} = \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} \left[-\frac{1}{x^2} \ln \left(\frac{1+x}{2+x} \right) + \frac{1}{x} \left(\frac{1}{1+x} - \frac{1}{2+x} \right) \right]$$

$$f'(1) = \frac{2}{3} \left[-\ln \frac{2}{3} + \frac{1}{6} \right] = \frac{2}{3} \ln \frac{3}{2} + \frac{1}{9}$$

$$f(-1) = g(-1) = -a = f'(1)$$

$$\therefore \frac{g(x)}{x} = a = -\frac{2}{3} \ln \frac{3}{2} - \frac{1}{9}$$

$$3. \quad (x+1)(x+2) \dots (x+n) = A_0 + A_1 x + \dots + A_n x^n$$

Put $x=1, 2, 3, \dots, (n+1)$ we have

$$(n+1)! = A_0 + A_1 + \dots + A_n$$

$$(x+1)(x+2) \dots (x+n) = A_0 + A_1 x + \dots + A_n x^n$$

Taking log we get

$$\log(x+1) + \log(1+2) \dots + \log(x+n) =$$

$$\log(A_0 + A_1 x + \dots + A_n x^n)$$

Diff. w.r.t. x

$$\frac{1}{x+1} + \frac{1}{x+2} + \dots + \frac{1}{x+n} =$$

$$\frac{A_1 + 2.A_2 + A_3(3x^2) \dots + A_n(nx^n)}{A_0 + A_1 x + \dots + A_n x^n}$$

Put $x=1$

$$\frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n+1} = \frac{A_1 + 2A_2 + \dots + nA_n}{(n+1)!}$$

$$4. \quad f(x) = \sqrt{(\sqrt{x-2} + \sqrt{2})^2} + \sqrt{(\sqrt{x-2} - \sqrt{2})^2}$$

$$= \sqrt{x-2} + \sqrt{2} + |\sqrt{x-2} - \sqrt{2}|$$

$$2 \leq x \leq 4 \Rightarrow f(x) = |\sqrt{x-2} - \sqrt{2}|$$

$$x > 4 \Rightarrow f(x) = \sqrt{x-2} + \sqrt{2} + \sqrt{x-2} - \sqrt{2} = 2\sqrt{x-2}$$

$$\therefore f'(x) = 0 \text{ in } (2, 4) \text{ is } \frac{1}{\sqrt{x-2}} \text{ in } (4, \infty)$$

$$\therefore f'(3) + f'(6) = 0 + \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$5. \quad x < -1 \Rightarrow f(x) = 2 - x + x - 1 + x + 1 = x + 2$$

$$-1 < x \leq 0 \Rightarrow f(x) = 2 - x + x - 1 - x - 1 = -x$$

$$0 < x \leq 1 \Rightarrow f(x) = 2 + x + x - 1 - x - 1 = x$$

$$1 < x \Rightarrow f(x) = 2 + x - x + 1 - x - 1 = 2 - x$$

$$y' \left(-\frac{1}{2} \right) + y' \left(\frac{1}{2} \right) + y' \left(\frac{3}{2} \right) + y' \left(\frac{5}{2} \right) = -1 + 1 - 1 - 1 = -2$$

$$6. \quad \lim_{x \rightarrow 0} \left(\frac{f(2-x^2)}{f(2)} \right)^{\frac{4}{x^2}} = 1^\infty \text{ form}$$

$$= e^{\lim_{x \rightarrow 0} \left(\frac{f(2-x^2) - f(2)}{x^2} \right) \cdot \frac{4}{f(2)}} = e^{f'(2) \cdot \frac{4}{f(2)}} = e^{\frac{-5.4}{40}} = e^{\frac{-1}{2}}$$

$$7. \quad y = \frac{\log(|\cos 4x| + |\sin x|)}{\log \sec 2x}$$

$$= \frac{3\sqrt{3}}{2 \ln(2)} = \frac{3\sqrt{3}}{\ln 4}$$

$$8. \quad \frac{f(x+y) - f(x)}{2} = \frac{f(y) - a}{2} + xy$$

$$f(x+y) = f(x) + f(y) - a + 2xy$$

$$\text{Put } x=y=0 \Rightarrow f(0) = a$$

$$f^1(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) + f(h) - a - 2xh - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - a}{h} + 2x$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + 2x = f^1(0) + 2x$$

$$f^1(x) = 2x + \sqrt{5a-1-a^2}$$

$$\text{Integrating } f(x) = x^2 + \sqrt{5a-1-a^2}x + c$$

$$\text{Putting } x=0 \Rightarrow c=a$$

$$\therefore f(x) = x^2 + \sqrt{5a+1-a^2}x + a$$

$$= \left(x + \frac{\sqrt{5a-1-a^2}}{2} \right)^2 + \frac{a^2 - a + 1}{n} > 0$$

$$f(x) > 0 \quad \forall x \in R$$

$$9. \quad \frac{dy}{dx} = \frac{1}{\frac{dy}{dx}}$$

$$\frac{d^2x}{dy^2} = \frac{d}{dy} \left(\frac{1}{\frac{dy}{dx}} \right) = \frac{d}{dx} \left(\frac{1}{\frac{dy}{dx}} \right) \times \frac{dx}{dy}$$

$$= \frac{-1}{\left(\frac{dy}{dx} \right)^2} \frac{d^2y}{dx^2} \cdot \frac{dx}{dy}$$

$$\frac{d^2x}{dy^2} = -\frac{1}{\left(\frac{dy}{dx} \right)^3} \times \frac{d^2y}{dx^2}$$

$$\frac{dy}{dx} = \left(\frac{-\frac{d^2y}{dx^2}}{\frac{d^2x}{dy^2}} \right)^{\frac{1}{3}}$$

$$1 + \left(\frac{dy}{dx} \right)^2 = 1 + \left(\frac{-\frac{d^2y}{dx^2}}{\frac{d^2x}{dy^2}} \right)^{\frac{2}{3}}$$

$$\text{So } R^{2/3} = \frac{1 + \left(\frac{dy}{dx} \right)^2}{\left(\frac{d^2y}{dx^2} \right)^{\frac{2}{3}}}$$

$$= \frac{1}{\left(\frac{d^2y}{dx^2} \right)^{\frac{2}{3}}} + \frac{\left(\frac{d^2y}{dx^2} \right)^{\frac{2}{3}}}{\left(\frac{d^2y}{dx^2} \right)^{\frac{2}{3}} \left(\frac{d^2x}{dy^2} \right)^{\frac{2}{3}}}$$

MULTI ANSWER QUESTIONS

$$10. \quad f(x) = \begin{cases} \pi - x, & \text{in } \left(\frac{\pi}{2}, \frac{3\pi}{2} \right) \\ x - 2\pi, & \text{in } \left(\frac{3\pi}{2}, \frac{5\pi}{2} \right) \end{cases}$$

$$11. \quad y = \frac{1 - \sqrt{1 - t^4}}{t^2}, \text{ Put } t^2 = \sin \theta$$

$$\Rightarrow y = \frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$$

$$\therefore \frac{dy}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} \text{ and } x = \cos \theta$$

$$\Rightarrow \frac{dx}{d\theta} = -\sin \theta$$

STATEMENT TYPE QUESTIONS

12. Both the statements are true, but statement 2 is not correct explanation of statement 1.

Statement 1 is true as period of $\sin x$ is 2π

Or, in general if for $y = f(x)$, $f(a) = f(b)$ we cannot say $f'(a) = f'(b)$

13. Given $f(x + y^3) = f(x) + f(y^3) \quad \forall x, y \in \mathbb{R}$

Put $x = y = 0$, we get

$$f(0 + 0) = f(0) + f(0) \Rightarrow f(0) = 0$$

now, put $y = -x^{1/3}$ we get $f(0) = f(x) + f(-x)$

$$\Rightarrow f(x) + f(-x) = 0$$

$\Rightarrow f(x)$ is an odd function

$\Rightarrow f'(x)$ is an even function

$$\Rightarrow f(-2) = a$$

COMPREHENSION QUESTIONS

14. From the given information, we have

$$f(x) = (x - c)^m g(x)$$

Where $g(x)$ is polynomial of degree $(n - m)$

Then $x = c$ is common root for the equations

$$f(x) = 0, f^1(x) = 0, f^2(x) = 0, \dots, f^{m-1}(x) = 0$$

Where $f^r(x)$ represent r th derivative of $f(x)$ w.r.t x .

15. Let $f(x) = a_1x^3 + b_1x^2 + c_1x + d_1 = 0$

has roots α, α, β then

$$g(x) = a_2x^3 + b_2x^2 + c_2x + d_2 = 0$$

must have roots α, α, β then

DIFFERENTIATION

$$g(x) = a_2x^3 + b_2x^2 + c_2x + d_2 = 0$$

must have roots α, α, β

$$\Rightarrow a_1\alpha^3 + b_1\alpha^2 + c_1\alpha + d_1 = 0 \text{ \& \dots\dots\dots(1)}$$

$$\Rightarrow a_2\alpha^3 + b_2\alpha^2 + c_2\alpha + d_2 = 0 \text{ \dots\dots\dots (2)}$$

α is also a root of equations

$$f'(x) = 3a_1x^2 + 2b_1x + c_1 = 0$$

$$\Rightarrow 3a_1\alpha^2 + 2b_1\alpha + c_1 = 0 \text{ \& \dots\dots\dots (3)}$$

$$3a_2\alpha^2 + 2b_2\alpha + c_2 = 0 \text{ \dots\dots\dots (4)}$$

Also from $a_2(1) - a_1(2)$ we have

$$(a_2b_1 - a_1b_2)\alpha^2 + (c_1a_2 - c_2a_1)\alpha + d_1a_2 - d_2a_1 = 0 \text{ \dots\dots\dots (5)}$$

Eliminating α from (3), (4) and (5) we have

$$\begin{vmatrix} 3a_1 & 2b_1 & c_1 \\ 3a_2 & 2b_2 & c_2 \\ a_2b_1 - a_1b_2 & c_1a_2 - c_2a_1 & d_1a_2 - d_2a_1 \end{vmatrix} = 0$$

16. Concept of multiple root

MATRIX- MATCHING QUESTIONS

17. a) We know that

$$2 \tan^{-1} x = \begin{cases} \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } -1 \leq x \leq 1 \\ \pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x > 1 \\ -\pi - \sin^{-1}\left(\frac{2x}{1+x^2}\right), & \text{if } x < -1 \end{cases}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{2}{1+x^2} \text{ if } x < -1 \text{ or } x > 1$$

$$b) \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \begin{cases} \tan^{-1} x, & x \geq 0 \\ -\tan^{-1} x, & x < 0 \end{cases}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{1+x^2} \text{ if } x < 0$$

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$$c) y = |e^{|x|} - e| = \begin{cases} |e^x - e|, & x \geq 0 \\ |e^{-x} - e|, & x < 0 \end{cases} = \begin{cases} e^x - e, & x \geq 1 \\ e - e^x, & 0 \leq x < 1 \\ e^{-x} - e, & x < -1 \end{cases}$$

$$\Rightarrow \frac{dy}{dx} > 0 \text{ if } x > 1 \text{ or } -1 < x < 0$$

$$d) u = \log|2x|, v = |\tan^{-1} x|$$

$$\Rightarrow \frac{du}{dx} = -\frac{1}{x}, \text{ and } \frac{dv}{dx} = \begin{cases} \frac{1}{1+x^2}, & x > 0 \\ -\frac{1}{1+x^2}, & x < 0 \end{cases}$$

$$\Rightarrow \frac{du}{dv} = \begin{cases} \frac{1+x^2}{x}, & x > 0 \\ -\frac{1+x^2}{x}, & x < 0 \end{cases}$$

Now we know that

$$\frac{1+x^2}{x} = x + \frac{1}{x} > 2 \text{ if } x > 1 \text{ and } < -2 \text{ if } x < -1$$

$$\Rightarrow \frac{du}{dv} > 2 \text{ if } x < -1 \text{ or } x > 1$$

INTEGER TYPE QUESTIONS

$$18. \begin{vmatrix} X & Y & Z \\ X_1 & Y_1 & Z_1 \\ X_2 & Y_2 & Z_2 \end{vmatrix} = \begin{vmatrix} X & SX & tX \\ 1 & S+XS_1 & t+Xt_1 \\ 0 & 2S_1+XS_2 & 2t_1+Xt_2 \end{vmatrix}$$

$$= X^2 \begin{vmatrix} 1 & S & t \\ 0+X & S_1 & t_1 \\ 0 & S_2 & t_2 \end{vmatrix}$$

$$= X^3 \begin{vmatrix} S_1 & t_1 \\ S_2 & t_2 \end{vmatrix} \Rightarrow K = 3$$

$$19. \ln f(x) = \sum_{n=1}^{100} n(101-n) \ln(x-n)$$

Differentiating w.r.t to x

$$\frac{f'(x)}{f(x)} = \sum_{n=1}^{100} \frac{n(101-n)}{(x-n)}$$

$$\Rightarrow \frac{f'(101)}{f(101)} = \sum_{n=1}^{100} \frac{n(101-n)}{(101-n)} = 5050$$

$$\Rightarrow \frac{f(101)}{f'(101)} = \frac{1}{5050} \Rightarrow k = 1$$

$$20. \begin{aligned} g(x+1) &= \log(f(x+1)) \\ &= \log x + \log f(x) \\ &= \log x + g(x) \\ \Rightarrow g(x+1) - g(x) &= \log x \end{aligned}$$

$$\Rightarrow g''(x+1) - g''(x) = -\frac{1}{x^2}$$

$$x \rightarrow x - \frac{1}{2}$$

$$\Rightarrow g''\left(x + \frac{1}{2}\right) - g''\left(x - \frac{1}{2}\right) = \frac{-4}{(2x-1)^2}$$

Put $x = 1, 2, 3, \dots, N$ and add

$$\Rightarrow g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right)$$

$$= -4 \left[1 + \frac{1}{9} + \frac{1}{25} + \dots + \frac{1}{(2N-1)^2} \right]$$

$$\Rightarrow k = 4$$

$$21. g'(y) = \frac{1}{\left(\frac{dy}{dx}\right)} = \frac{1}{2e^{\left(\frac{1-x}{2}\right)} + x^2 + x + 1}$$

$$\therefore y = -\frac{7}{6} \Rightarrow x = 1$$

$$g'\left(-\frac{7}{6}\right) = \frac{1}{5}, k = 5$$

$$22. \ln f(x) = \sum_{n=1}^n \ln(x-n)$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \sum_{n=1}^n \frac{1}{x-n}$$

$$\Rightarrow f'(x) = f(x) \sum_{n=1}^n \frac{1}{x-n}$$

$$\begin{aligned} \text{Put } x = n \Rightarrow 5040 &= (n-1)(n-2) \dots 3.2.1 \\ &= (n-1)! \Rightarrow n = 8 \end{aligned}$$

$$23. f(x) = x + \frac{1}{x + f(x)} \Rightarrow f^2(x) - x^2 = 1$$

$$\Rightarrow 2f(x) \cdot f'(x) - 2x = 0 \Rightarrow f(x) \cdot f'(x) = x$$

$$\Rightarrow f(50) \cdot f'(50) = 50$$

$$\Rightarrow \frac{f(50) \cdot f'(50)}{2} = 25$$

$$\Rightarrow \sqrt{\frac{f(50) \cdot f'(50)}{2}} = 5$$

$$24. 21 = 4 - 7 + k \Rightarrow k = 24 \Rightarrow \frac{k}{4} = 6$$

$$25. f'(x) \cdot f(-x) - f(x) \cdot f'(-x) = 0$$

$$\Rightarrow \frac{d}{dx}(f(x) \cdot f(-x)) = 0$$

$$26. z = \ln\left(\tan \frac{x}{2}\right) \Rightarrow \frac{dz}{dx} = \frac{1}{\sin x} = \operatorname{cosec} x$$

$$\therefore \frac{dy}{dx} = \frac{dy}{dz}(\operatorname{cosec} x)$$

$$\Rightarrow \frac{d^2 y}{dx^2} + 4y = 0 \Rightarrow K = 4$$

$$27. \frac{dy}{dx} = K \cdot x \cos kx + \sin kx \Rightarrow \frac{d^2 y}{dx^2}$$

$$= -k^2 x \sin kx + k \cos kx + k \cos kx$$

$\therefore$ given equation is

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$$2k \cos kx - k^2 x \sin kx + x \sin kx$$

$$' = 2k \cos kx \Rightarrow 1 - k^2 = 0 \Rightarrow k = \pm 1$$

$$28. \quad f(f^{-1}(x)) = x \Rightarrow (f^{-1}(2))' = \frac{1}{3} \Rightarrow k = 3$$

SUBJECTIVE TYPE QUESTIONS

$$29. \quad f(x) = \begin{cases} \frac{x-1}{2x^2-7x+5}, & x \neq 1 \\ \frac{1}{-3}, & x = 1 \end{cases} = \begin{cases} \frac{x-1}{(x-1)(2x-5)}, & x \neq 1 \\ \frac{1}{-3}, & x = 1 \end{cases}$$

$$= \begin{cases} \frac{1}{2x-5}, & x \neq 1 \\ -\frac{1}{3}, & x = 1 \end{cases}$$

$$\begin{aligned} f'(x)_{x=1} &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{1}{2(h+1)-5} + \frac{1}{3}}{h}, = \lim_{h \rightarrow 0} \frac{2h}{3h(2h-3)} \\ &= \lim_{h \rightarrow 0} \frac{2}{(2h-3) \cdot 3} = \frac{-2}{9} \end{aligned}$$

$$\begin{aligned} 30. \quad y &= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{c}{x-c} + 1 \\ \Rightarrow y &= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \left(\frac{c+x-c}{x-c} \right) \\ \Rightarrow y &= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx}{(x-b)(x-c)} + \frac{x}{x-c} \\ \Rightarrow y &= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{bx+x(x-b)}{(x-b)(x-c)} \\ \Rightarrow y &= \frac{ax^2}{(x-a)(x-b)(x-c)} + \frac{x^2}{(x-b)(x-c)} \end{aligned}$$

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$$\Rightarrow y = \frac{ax^2 + x^2(x-a)}{(x-a)(x-b)(x-c)}$$

$$\Rightarrow y = \frac{x^3}{(x-a)(x-b)(x-c)}$$

$$\Rightarrow \log y = \log \left\{ \frac{x^3}{(x-a)(x-b)(x-c)} \right\}$$

$$\Rightarrow \log y = 3 \log x - \{ \log(x-a) + \log(x-b) + \log(x-c) \}$$

On differentiating w.r.t x we get

$$\frac{1}{y} \frac{dy}{dx} = \frac{3}{x} - \left\{ \frac{1}{x-a} + \frac{1}{x-b} + \frac{1}{x-c} \right\}$$

$$\frac{dy}{dx} = y \left\{ \left(\frac{1}{x} - \frac{1}{x-a} \right) + \left(\frac{1}{x} - \frac{1}{x-b} \right) + \left(\frac{1}{x} - \frac{1}{x-c} \right) \right\}$$

$$\frac{dy}{dx} = y \left\{ \left(-\frac{a}{x(x-a)} - \frac{b}{x(x-b)} - \frac{c}{x(x-c)} \right) \right\}$$

$$\frac{dy}{dx} = \frac{y}{x} \left\{ \left(\frac{a}{a-x} + \frac{b}{b-x} + \frac{c}{c-x} \right) \right\}$$